### 2.1.2 The Electro-Hydraulic Actuation Chain

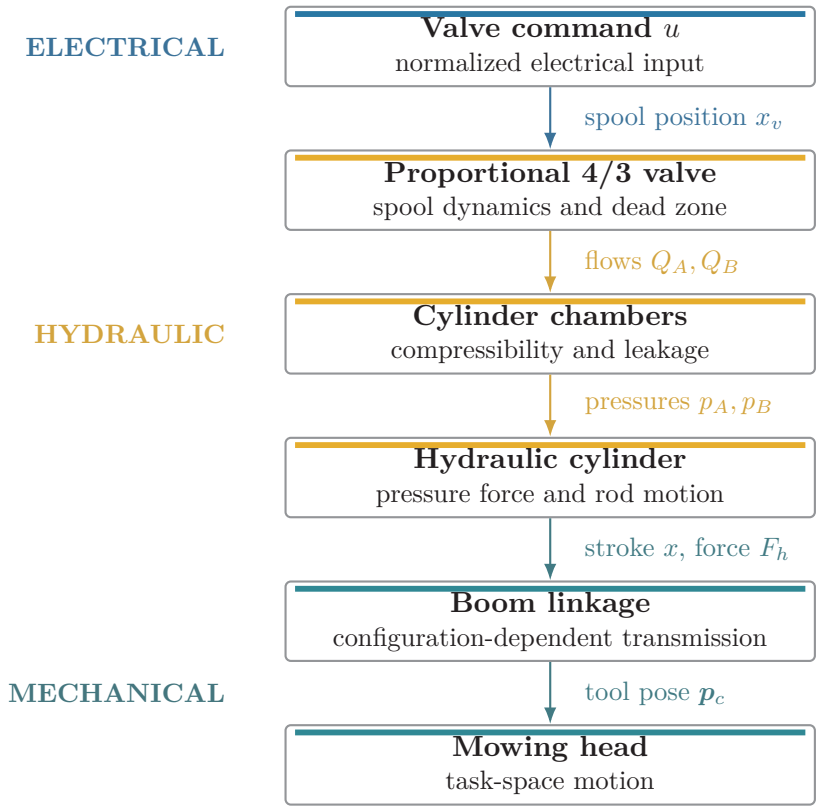



**Figure 2.1:** The electro-hydraulic actuation chain from valve command to mowing-head motion. The block colors distinguish the electrical, hydraulic and mechanical stages; arrow labels identify the signals passed between consecutive stages.

In Figure 2.1, an electrical command moves a proportional valve spool and meters flow to the cylinder chambers. The orifice law [Merritt 67] makes flow depend on spool position and the square root of the pressure margin. Valve authority is therefore load-dependent and directional; overlap creates a dead zone near null and the spool saturates at large commands [Jelali 03].

Compressibility, changing chamber volume, relief limits, leakage and seal friction further separate flow from motion. Several of these effects are non-smooth, while bulk modulus leakage and friction drift with temperature and wear. The mechanical linkage then converts cylinder translation through configuration-dependent lever arms. Chapter 5 consequently models the hydraulic realization in cylinder space, before kinematic effects are mixed into cutter error.

### 2.1.3 High-Fidelity Simulation[1] as a Data Source

The component-level Simscape plant [The MathWorks, Inc. 24a] resolves the stiff hydraulic dynamics but runs one machine at small integration steps. Policy optimization instead benefits from thousands of GPU-parallel plants [Makoviychuk 21]. The detailed model is therefore used offline to generate identification data; compact actuator models carry its behavior into parallel training, as summarized in Figure 2.2.